

Energy Modelling

Renewable Energy 80% by 2030

Future Supply

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Brief

Aim

Refine the assumptions and modelling of a key piece of intelligence provided by external consultants *Aurora* in their report *Accelerating Renewables in Northern Ireland, A high level design of a Support Scheme*. Their report details projections of renewable projects development through to completion out to 2030. There is criticism of the assumptions used around pipeline progression and risk weighting viability of projects in the current pipeline. There is no clear awareness of projects in the pre-planning phase, yet to submit for planning permission. There is subtle nuance that we will make explicit, between probable and realistic projections to 2030, and variables needed to ensure targets by 2030.

To add value our work should borrow the bulk of correct work in the report, verify with experts and more clearly present figures.

Background

The climate change act contains the stipulation that Northern Ireland meet 80% of total energy supply from renewable energy sources (RES) by 2030.

Target viability and trajectory require

1) Determination of **current renewable energy generation**. This is the current, existing, renewable generation which we assume to be constant to 2030.

2) The **pipeline of renewable energy projects forthcoming**, and their generation capacity coming available over the next 5 years from 2025 - 2030.

3) Estimation or determination of the **Total Energy Requirement (TER)**, from which we can take 80% as the absolute target figure.

A corollary of this is that fossil fuel and other non - renewable supply are elastic enough to scale down on the basis of increased renewable supply, thus the pipeline of currently developing non-renewable sources are beyond our scope.

5 years is noted to be a short time frame on the scale of energy and specifically renewable project development. A more complete consideration arising from this modelling work would involve future energy capacity estimation from projects that may be anticipated to start planning soon, based on current trends of project application and development.

It is noted in commentary that the current trajectory is way off the expected trajectory and that the target is unrealistic both with the current pipeline of project capacity at the planning stage or later, and absent any real advance in offshore wind capacity.

Demand and Supply

Supply is obviously designed to meet demand. The difference between generation and consumption is more commonly discussed.

Generation is total generated power in the Country.

Consumption is net of transmission and distribution losses, own use, energy exports and other power not consumed in NI, but is inclusive of energy imports.

Non-Renewables

Assumed to be elastic and required only for demand renewables cannot meet. The principle of *priority dispatch* ensures that when renewable sources are available, they are used over non-renewable source.

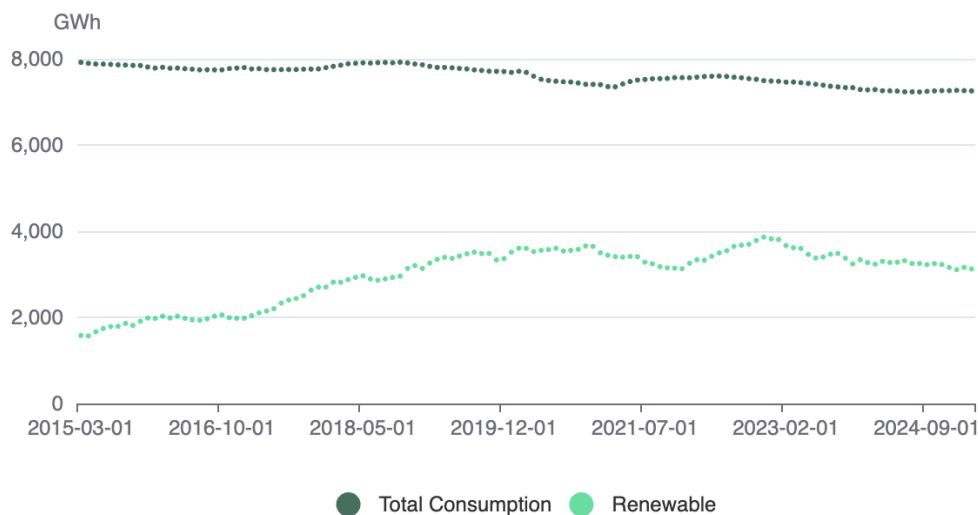
Renewables

Determining RES is the thrust of our modelling work. Future estimates are not well known.

Past trends of supply are presented below. Note the plateau in renewable sources and decreasing demand curves.

12 month Electricity Consumption and Renewable Generation (2015–2025)

Rolling 12 month volume to March 2025



The final reported 12 month period is further broken down by renewable type.

Renewable Energy breakdown

Generati on Type	Percentage Contribution	Volume Generated (GWhr)	Volume Generated (TWhr)	Capacity Computed (MW) *
Biogas	6.8	214	0.2	
Biomass	5.2	161	0.1	
Landfill gas	1.6	49	0.05	
Other	0.9	29	0.03	
Solar PV	3.6	112	0.1	
Wind	81.9	2,561	2.6	
Total	100	3,126	3.1	

April 2024 to March 2025

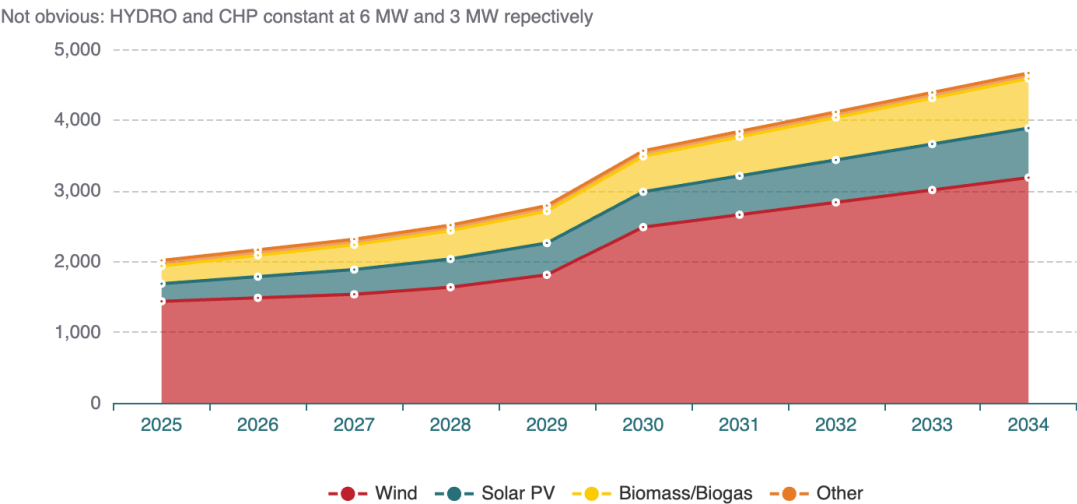
* CF = 22%

Future Generation and Consumption

SONI and EirGrids All Island Resource Adequacy Assessment gives a project of renewables broken down by type that goes out to 2034, including offshore wind, that miss the 80% by 2030 target.

The plot below is of projections from the All Ireland Adequacy Assessment. It gives forward estimates of renewable supply from 2025 into the future.
The methodology for projection is not released.

Renewable Energy Capacity Trends by Source



Key dates are below. There are notable differences in figure that we note and try and reconcile.

AIEAA projected All - Renewable sources

Year	RES (MW)	RES (TWhr) *	RES (TWhr) **	NISRA (TWhr)
2023				
2024				
2025	2,023	3.9	3.1	3.9
2030	3,573	6.9	5.5	
2034	4,673	9.0	7.2	

EirGrid and SONI | AIRAA | 2025-2034

* CF = 22% **CF = 17.5%

Discrepancy

Notes have been made of possible reasons for the discrepancy.

22% Capacity Factor is a large estimate, ~18% worked better

Mismatch between counted sources and the timings 2024-25 financial year vs. 2025 calendar year.

Measured vs. Estimated

The GWh figure may be net of outages, curtailment, own-use or exports, and not gross energy

They may be using average capacity over the year rather than nameplate.

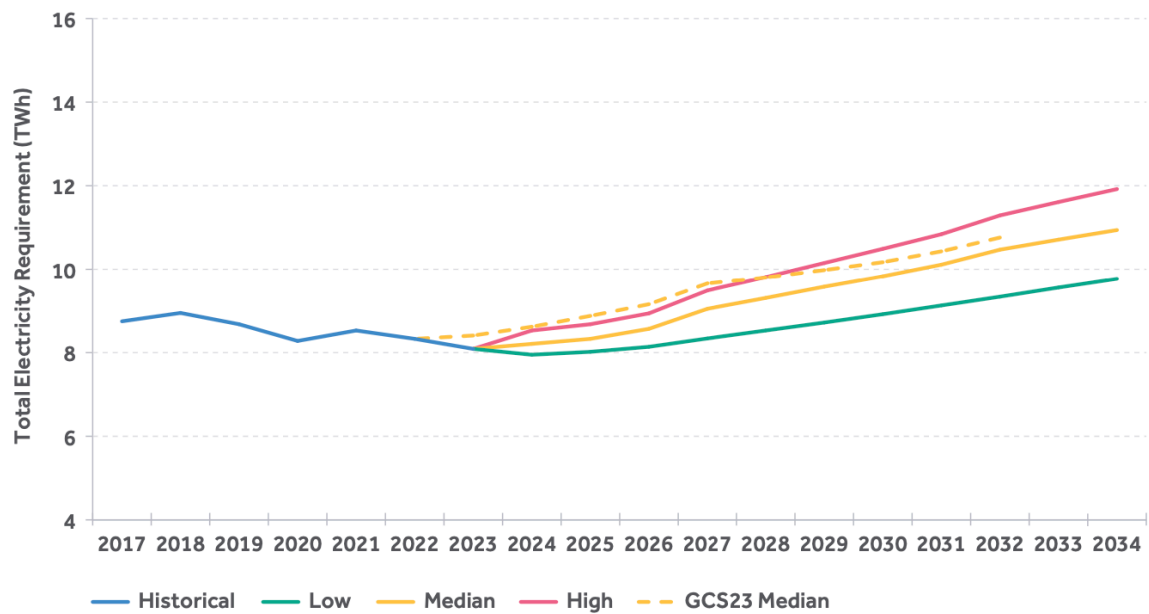
Nameplate capacity means the maximum rated output under optimal conditions. Its measured in Watts (GW/MW/TW). It's an instantaneous power measure, not energy over time.

Timing / commissioning profile may not be treated in same across sources, if applied.

Future Demand

Total Energy Requirement is lifted from SONI working paper. It details demand for Northern Ireland out to 2034.

Total Projected Electricity Demand Annually in Northern Ireland



Total electricity requirement for Northern Ireland

They explain the method somewhat. It is a linear regression across a number of economic forecasts, temperature and new anticipated technological load e.g. data centres.

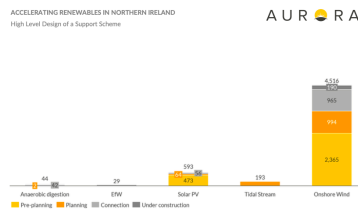
Requirement

	Renewable	Estimate	TER	Estimate	Percentage	Notes
2025	3.126 TWhr	Reported from NISRA	7.262 TWhr	Actual	43%	Current
2030	5.5 TWhr	Our adjusted generation from SONI CF:17.5%	10.2 TWhr	Median scenario	54%	Target Miss

Research and Literature

Aurora Borrows heavily from a RenewableNI KPMG report

Lineage



RenewableNI

Aurora

DfE

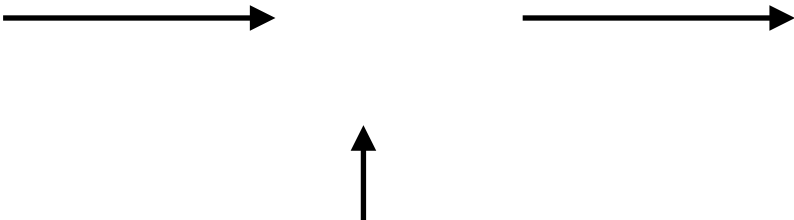


Table B3.4: Partially onshore dispatchable plant in Northern Ireland (MW)

At year end:	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
Large Scale Wind	1305	1305	1325	1405	1630	2305	2480	2625	2800	3005
Small Scale Wind	180	180	180	180	180	180	180	180	180	180
Solar PV	250	300	550	400	450	500	550	600	650	700
Small Scale Biogas	24	24	24	24	24	24	24	24	24	24
Landfill Gas	16	16	16	16	16	16	16	16	16	16
Small Scale Biomass	6	6	6	6	6	6	6	6	6	6
Renewable CHP	3	3	3	3	3	3	3	3	3	3
Other CHP	6	6	6	6	6	6	6	6	6	6
Small Scale Hydrop	6	6	6	6	6	6	6	6	6	6
Waste-to-Energy	15	15	15	15	15	15	15	15	15	15
Total	1765	1865	1965	2111	2336	3065	3286	3511	3736	3961

All Island Resource
Adequacy Assessment
Forecasts of Total Electricity
Requirement and renewables

RenewableNI Report

This report lays the groundwork for the Aurora Report, which borrows heavily from it.

The renewableNI - KPMG report sets out a **primary route to market**, mainly the Contract for difference (CfD) scheme and notes that it has been 7 years since a renewable support scheme has been available in Northern Ireland, and lack of newly commissioned renewables sources reflects the lack of support.

Only ~30 MW of new capacity was added in the past year (2022-23), compared to 400 MW/year at the 2016 peak.

- Without intervention, NI will fall far short of 2030 targets — best-case under “business as usual” is **725 MW new capacity by 2030**, less than 40% of what’s needed.
- These figures include estimates of projects in the pre-planning stage
- Grid, planning, and route-to-market issues are “single points of failure.”

They describe baseline project development status

Outlines the stages of projects’ development and their aggregate generation and capacity, by stage and technology. They also note projects in the *pre-planning* stage, not in the REPD database.

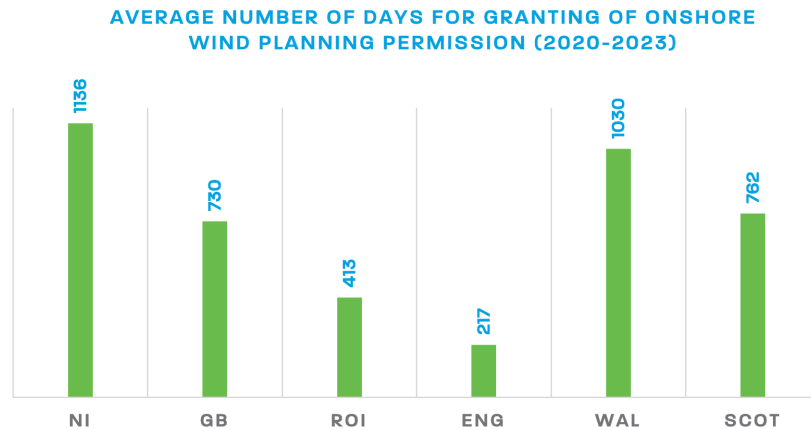
“ What is apparent from this data is that it is highly unlikely the additional capacity required to achieve 2030 targets will be delivered from the existing pipeline”

They risk weight the success probability of projects in each stage reaching completion.

They describe planning, connection and construction lags

They benchmark project development timelines, giving estimates of the planning, connection and construction phases **e.g. see the figure below.**

Figure 3



Projections

BAU Development Pathway

- Planning - 3 years
- Grid application (6 months) and connection (18 months) - 2 years
- Construction (18 - 24 months) - 2 years
- Generation would only begin in 2030 (report is dated from 2023)
- Capacity would only be **725 MW**

Optimised Pathway

- Planning - 1 years
- Grid application (6 months) and connection (18 months) - 2 years
 - OR Grid application (6 months) and expedited connection (6 months) - 1 year
 - OR **Concurrent** Planning permission with Grid application (6 months) and expedited connection (6 months) - 0 years
- Construction (18 - 24 months) - 2 years - no change from BAU
- Credible path to 2030 target
- New capacity could be **1,600 MW**

See the figures used as inputs below

See detailed breakdown below

Onshore Wind	Capacity (>5MW)	Capacity (1-5MW)	Capacity (<1MW)	Total Capacity (MW)	Additional capacity required by 2030 (MW)
Capacity	1577.25	43.8	149.49	1770.54	1000

Status	Capacity (>5MW)	Capacity (1-5MW)	Capacity (<1MW)	Total Capacity (MW)	Probability Weighting (business as usual)	Total Capacity delivered (MW)	Probability Weighting (optimised)	Total Capacity delivered (MW)
Under Construction	56.75	0	1.225	57.975	100%	57.98	100%	57.975
Consented with grid	186.3	0	0	186.3	80%	149.04	90%	167.67
Consented (no grid offer/unknown)	204.5	32.8	118.69	355.99	70%	249.19	80%	284.792
In planning	262.4	11	29.025	302.425	60%	181.46	70%	211.6975
Pre Planning	867.3	0	0.55	867.95	20%	173.59	40%	347.18
Total	1577.25	43.8	149.49	1770.54		811.25		1069.31

Solar	Capacity (>5MW)	Capacity (1-5MW)	Capacity (<1MW)	Total Capacity (MW)	Additional capacity required by 2030 (MW)
Capacity	673.6	16.85	0.2	690.7	400

Status	Capacity (>5MW)	Capacity (1-5MW)	Capacity (<1MW)	Total Capacity (MW)	Probability Weighting (business as usual)	Total Capacity delivered (MW)	Probability Weighting (optimised)	Total Capacity delivered (MW)
Under Construction	0	0	0	0	100%	0	100%	0
Consented	64.5	7.1	0	71.6	75%	53.7	90%	64.44
In planning	50	4.75	0.2	54.95	60%	32.97	70%	38.465
Pre Planning	559.1	5	0	564.1	20%	112.82	40%	225.64
Total	673.6	16.85	0.2	690.65		199.49		328.55

Offshore Wind	Capacity (>5MW)	Capacity (1-5MW)	Capacity (<1MW)	Total Capacity (MW)	Additional capacity required by 2030 (MW)
Capacity	1000	0	0	1000	500

	Capacity (>5MW)	Capacity (1-5MW)	Capacity (<1MW)	Total Capacity (MW)	Probability Weighting (business as usual)	Total Capacity delivered (MW)	Probability Weighting (optimised)	Total Capacity delivered (MW)
Pre Planning	1000	0	0	1000	10%	100	20%	200

In undertaking our delivery timeline assessment, we have used the following headline assumptions.

Onshore Wind	Probability Weighting (business as usual)	Assumptions	Probability Weighting (optimised)	Assumptions
Under Construction	100%	All projects currently under construction will be delivered and operational by 2030.	100%	All projects currently under construction will be delivered and operational by 2030.
Consented with grid	80%	20% of consented projects with grid will not be delivered on time due to route to market or grid delays.	90%	Reduced route to market risk with introduction of government backed subsidy scheme.
Consented (no grid offer/unknown)	70%	We have assumed a further 10% reduction in success probability to account for grid offer uncertainty.	80%	10% reduction in probability to account for grid offer uncertainty.
In planning	60%	DfI Renewable energy planning applications list, April 2017 (post NIRO) to March 2023, 63% of all renewable applications have been approved (246 / 391). The remainder have been rejected, undecided or withdrawn.	70%	10% increase in probability weighting due to reduced decision timelines, increased planning officer education and strict consultation parameters.
Pre Planning	20%	Fairly low success probability assumed due to early stage nature of the assets, with planning, grid and route to market risk all remaining high.	40%	Higher success rate assumed due to improved planning resources, grid investment and route to market options.

Solar	Probability Weighting (business as usual)	Assumptions	Probability Weighting (optimised)	Assumptions
Under Construction	100%	All projects currently under construction will be delivered and operational by 2030.	100%	All projects currently under construction will be delivered and operational by 2030.
Consented	75%	Same methodology as Onshore Wind but a blended rate has been applied due to lack of information on grid connection status.	90%	Reduced route to market risk with introduction of government backed subsidy scheme.
In planning	60%	DfI Renewable energy planning applications list, April 2017 (post NIRO) to March 2023, 63% of all renewable applications have been approved (246 / 391). The remainder have been rejected, undecided or withdrawn.	70%	10% increase in probability weighting due to reduced decision timelines, increased planning officer education and strict consultation parameters.
Pre Planning	20%	Fairly low success probability assumed due to early stage nature of the assets, with planning, grid and route to market risk all remaining high.	40%	Higher success rate assumed due to improved planning resources, grid investment and route to market options.

Offshore Wind	Probability Weighting (business as usual)	Assumptions	Probability Weighting (optimised)	Assumptions
Total	10%	Fairly low success probability assumed due to early stage nature of the assets, with planning, grid and route to market risk all remaining high.	20%	Higher success rate assumed due to improved planning resources, grid investment and route to market options

RenewableNI - Detailed Scenario Modelling and Discussion

The report is dated, figures are stated for and scenarios estimated from : 2023

Business as Usual: 725 MW new capacity by 2030 → <40% to 80% target.

Optimised Pathway (if recommendations adopted): ~1,600 MW from existing pipeline + scope for additional projects → credible path to 80% target.

Summary of Report

According to the 'Shaping Our Electricity Future Roadmap Version 1.1' published by EirGrid and SONI in July 2023, there is a need for Northern Ireland to have built a renewable generation capacity of **3,550 MW by 2030 in order to deliver the target** of 80% of electricity consumption from renewables sources.

Analysis suggests that to achieve this, **1,900MW** of new generation from onshore wind, solar and offshore wind will need to be added by 2030.

The NI market has a known pipeline of **3,461MW** of incremental renewables, from pre-planning to consented assets.

Assuming current timelines, and applying probability weightings, a best-case outcome of **725MW** could commence generation by 2030. This is less than 40% of the volume required.

Furthermore, given the current timelines to progress a project from ideation to operation in NI, unless a project is already within the pipeline, it is highly unlikely to be able to contribute to the 2030 target.

This is outlined in the "Business as Usual" scenario.

The same pipeline of 3,461MW, but adopting and delivering upon the recommendations contained within this report, we estimate that the same pipeline could deliver **1,600 MW** of generation capacity by 2030.

Furthermore, the enhanced timelines would provide an opportunity for **further projects** to be introduced into the system and still connect by 2030,

RenewableNI figures reported

3,550 MW

Total RES to meet
80% by 2030

3,461MW

Known Pipeline

1,900 MW

New RES needed
added
for 80% by 2030

- Implies current RES baseline capacity is 1,650 MW
- Assumes this baseline RES capacity stays until 2030
- BAU pathway can return 750MW
- Optimised path can return 1,600 MW

Aurora Report

Commentary

Section 1.5 on page 17

1.5 Renewables pipeline in Northern Ireland

Since the closure of the Northern Ireland Renewable Obligation (NIRO) scheme for new entrants in 2017, no significant renewables capacity has become operational in Northern Ireland.

The renewables pipeline in Northern Ireland is dominated by onshore wind. There is good visibility on the capacity of plants having submitted planning applications, as they are listed in the UK government's Renewable Energy Planning Database (REPD). Onshore wind accounts for around 900 MW (76%) of capacity currently in the planning stage or beyond (grid connection and construction). Solar PV accounts for 13% with around 150 MW.

Offshore - targets from DfE of 1GW by 2030 are ambitious and not generally included.

The drivers of growth are **onshore wind and photovoltaics**

Construction Pipeline Progression

Success probability

100% progression from stage to stage

But Projects are not always successfully in auction participation. They phrase this from the auctions context frame, but describing Win Rates (percentage of total auction capacity won)

Time to Event of Pipeline Progression in BAU

Planning 3 years

Grid connection 2 years

Construction 1 year

Cumulative 6 years from planning app to Completion

Necessary Cadence of Progression to meet 80% by 2030

Planning 1 year

Connection 2 years

Construction 1 year

Pipeline distribution (end-2023 snapshot)

By stage share of annual GWh in pipeline:

Pre-planning 53%

Planning 23%

Connection 20%

Under construction 4%

Absolute RES Target

Aurora reference the AIRAA median demand forecast to 2030 and take 80% of this 10.2 TWh as the target.

Of this ~8.4 TWh, 3.4TWh generation of renewables exist and are not going anywhere.

Of the remaining 5TWh they estimate 1.5TWh (~30%) are able to proceed without support from any scheme.

This leaves 3.5 TWhr needed to be delivered by 2030.

Auction Design

Auctions are explained in the glossary in the appendix of the document. Any economic and capacity constraints are not considered although the win rate is.

Pot sizing and technology is not considered.

Aurora anticipate two auctions that impact the availability of new generation by 2030. The total sum of which is the 3.5GWh generation needed.

Auction 1: 2025 (delivery 2027) – 1,000 GWh/y ($\approx$ 500 MW).

Auction 2: 2027 (delivery 2029) – 2,500 GWh/y ($\approx$ 1,250 MW).

These are the **procurement ratios** Aurora says are needed

The “necessary conditions” case

Planning takes 1 year:

Auction 1 (2025): 48% of eligible volume of Projects must win.

Auction 2 (2027): 74% of eligible volume of Projects must win.

Planning takes 2 years

Auction 1 (2025): 85% must win (vs. 48%).

Auction 2 (2027): 92% must win (vs. 74%).

Other Assumptions Aurora specify

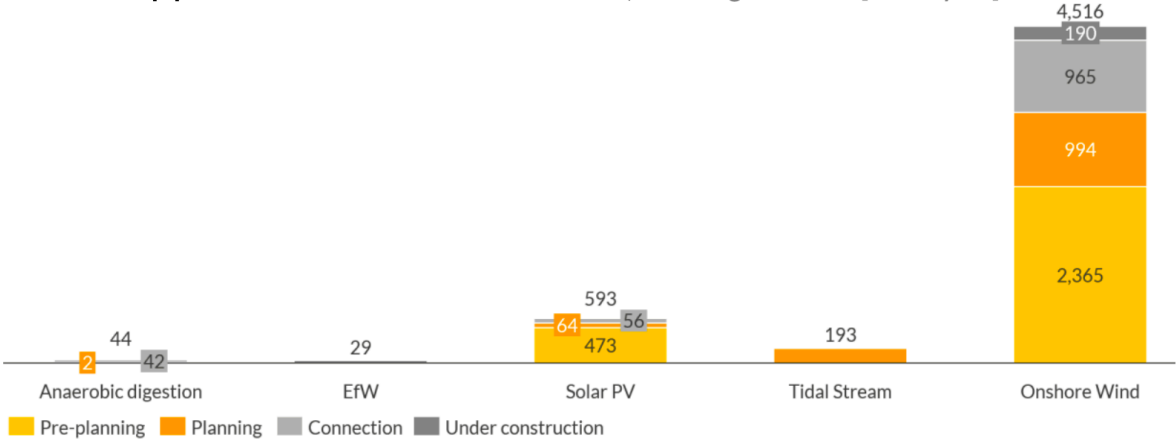
All capacity procured in Auctions 1 and 2 becomes operational on schedule (max 2 years after auction)

Projects delay construction to remain auction-eligible (e.g., 2024-eligible wait for 2025 auction, unsuccessful 2025 + 2026-eligible wait for 2027).

Pre-planning inflow assumed: extra 1,000 MW onshore + 600 MW solar; 50% get planning in 2026, 50% in 2027. This is in addition as to what is register in REPD and all of which will receive planning permission by the 2027 auction.

Current Pipeline

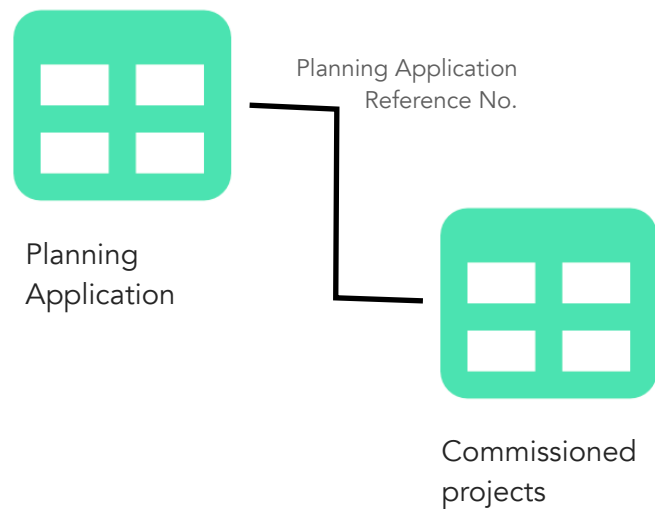
Renewables pipeline of Northern Ireland in late 2023, Annual generation [GWh/year]



Sources: DESNZ REPD (planning stage and beyond); RenewableNI (pre-planning). See below for mapping reported project status to one of planning, connection, construction.

Our Modelling

Tracking projects across datasets



The NI Renewable Energy Planning Database (REPD) gives details of project planning applications and their status, whether, pending, granted, denied.

UK REPD gives Northern Ireland projects as well, but provides the fuller timeline of project development with timestamped dates of project submission, permission granted, construction started and project operational.

Combining the two we found all the projects in NI referenced dataset where present in the UK dataset. As it gave the requisite detail, we continued to use the the UK dataset in isolation.

Categorising and Filtering Projects

We analysed the UK REPD to derive estimates of

1. Project stage success rates
2. Development time of project progression from stage to stage
3. Determine the current state of the project development pipeline

Pipeline Status

We categorised viable projects as in the pipeline stages below, mapped to their assigned 'status'

c('Abandoned',

'Appeal Refused',
'Appeal Withdrawn',
'Application Refused',
'Application Withdrawn',
'Planning Permission Expired',
"Revised") ~ 'Failed',

`Development Status (short)` %in%
c("Application Submitted" ,
"Appeal Lodged") ~ 'Planning',

`Development Status (short)` %in%
c("Planning Permission granted",
"Appeal granted",
"Secretary of state granted",
'Awaiting Construction'
) ~ 'Connection',

`Development Status (short)` %in%
c('Construction',
'Under Construction'
) ~ 'Construction',

roughly as per Aurora criteria below

To Note Aurora map to slightly different naming conventions, e.g. 'Under Construction' instead of Construction as it was in our datasets. As is some naming convention included that we mapped ourselves above

Also note that the Revised was moved from 'planning' status to 'failed'. An important distinction is the grain, or primary key of the database is the Application - not a project. To say an application has 'failed' is not the same as the declaration a project has failed. We adopted this mapping as it was observed 'Revised' applications almost always made a new application, with a new reference and unique identifier as another row in the database. We track the success of this instead.

Appeals are attached to the same application reference and kept in the same row so do not apply to this case.

Due to uncertainty in the application of the database maintainers in the project status and discrepancy with status and timestamped progression. We introduced an additional ruleset below to enforce some common sense interpretation on perceived data inconsistencies

Those projects that have a planning permission date expired up to 2 years ago are excluded. Many of these projects were observed to still have a status of 'Awaiting Construction'. Planning expires about 5 years from grant. The leeway is that some projects have a planning permission

expired date, but have proceeded to construction. It may reflect a delay in updating the dataset on the part of the maintainers.

2) Those projects that have the '**Planning Submitted**' date more than 5 years in the past are marked as failed. Many of these are marked as '**Planning Permission Submitted**'. It is taken as unreasonable that even with appeals, that these are actively being pursued.

For the purpose of fixing the time for consistent analysis 'today' refers to
 Sys.Date()
 # "2025-08-31"

Planning_expired < as.Date("2025-08-31") - 2*365

Planning < as.Date("2025-08-31") - 5*365

Process step	Description	Typical duration (years)
Planning	- Planning spans the period between a planning application being submitted and planning permission being granted - Includes all projects with a status of 'planning application submitted', 'appeal lodged' and 'revised' in the REPD	1 – 3
Connection	- Connection spans the period from obtaining a grid connection to the beginning of the construction process - Includes all projects with a status of 'planning permission granted', 'appeal granted', and 'secretary of state granted' in the REPD	1 – 2
Construction	- Connection spans the period from the start of construction to the COD and includes the period of financing the project - Includes all projects with a status of under 'construction'	1

Commissioned projects

Further Completed Projects (and not considered in the pipeline) :

```
`Development Status (short)` %in%
  c('Decommissioned', #no NI relevant returns
    'Operational') ~ 'Completed')
```

This gives our best guess at the pipeline of projects in development currently

Missing Capacity Estimates

Not all projects had Capacity estimates attached in the REPD. Upon inspection these were found to be either updates to projects, or amendments not reflected in a revision or other reapplication.

It is sufficient to expect that capacity is included in figures for current renewable energy sources.

We do not estimate capacity in this case for those projects applications that are missing capacity figures.

Further, when estimating pre-planning application submissions from the current database we should remove these from training and sampling data as they are indicative of updates rather than new capacity.

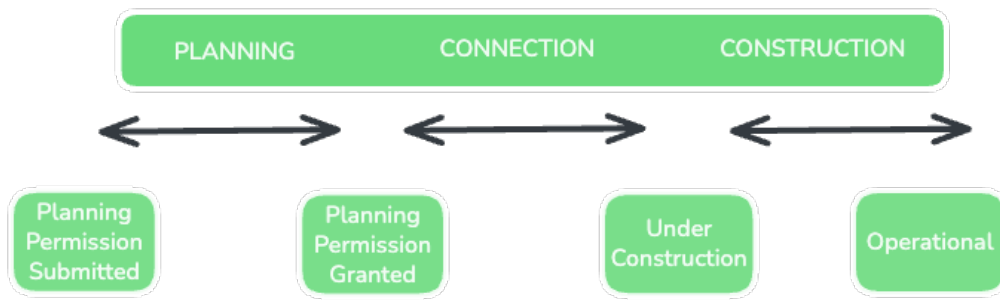
Project development timelines

Project Progress is timestamped in the UK REPD database. This provides the opportunity to derive time between discrete events. See our results below and our scheme and definitions further down again. We derive results from any project that completed that phase regardless of whether it has been completed and operational, whether it is still in the pipeline, or whether it was subsequently abandoned further down the pipeline.



Note: The scales on the x and y axis are free and not consistent

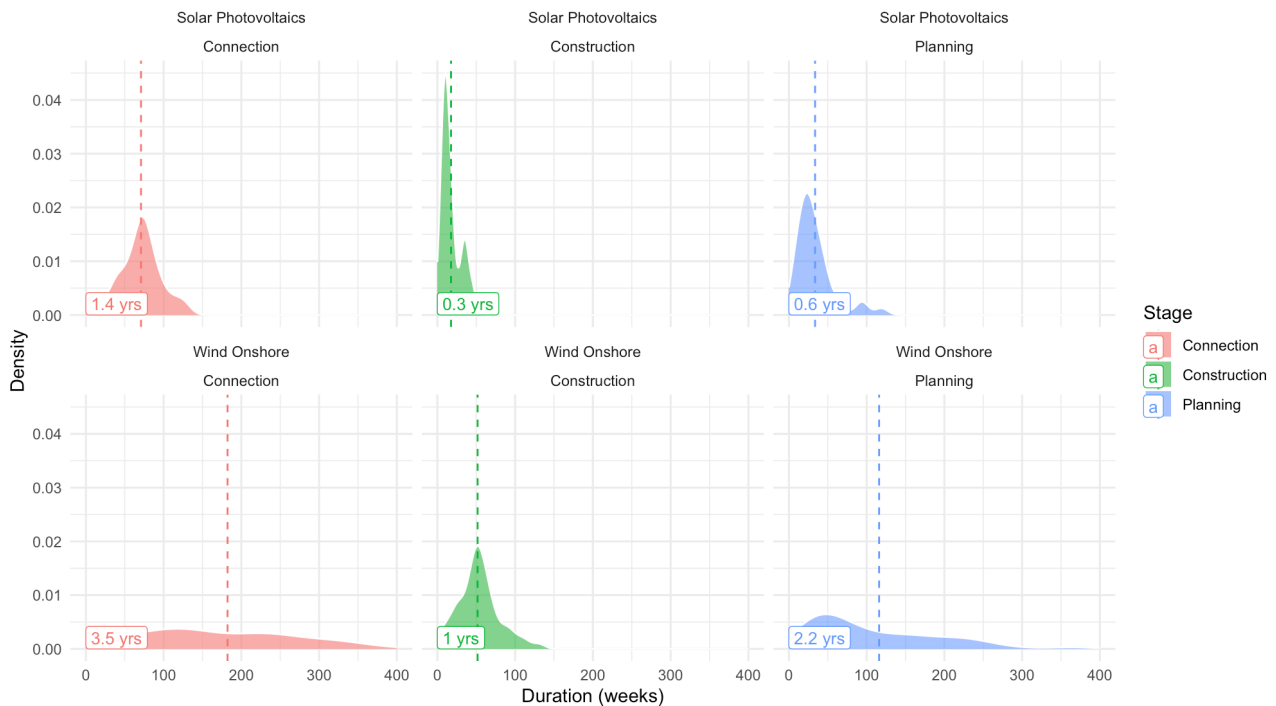
Planning = `Planning Permission Granted` - `Planning Application Submitted`
Connection = `Under Construction` - `Planning Permission Granted`
Construction = Operational - `Under Construction`



CONNECTION

Assumed to occupy the time between planning granted and construction starting

Time from Submission to Planning Granted, Connection finished and Operation (Construction Finished)
Includes all projects that have completed a stage regardless where they subsequently dropped out, or are still in the pipeline



Note: The scales on the x and y axis are fixed for comparison

The same graph but with consistent x and y axis for comparison

Project development stage drop out rates

We look at projects and note when a recorded `status`, or our enforced expiry ruleset, has indicated a project has not proceeded to the next stage.

#	Current Status	No Planning Granted Date (Planning phase)	No Construction Started Date (Connection Phase end)	No Operational Date (Construction Phase end)	N	Capacity (MW)	# NA Capacity
1	Completed	FALSE	FALSE	FALSE	78	1083.3	0
2	Completed	FALSE	FALSE	TRUE	1	23.5	0
3	Completed	FALSE	TRUE	FALSE	22	279.3	0
4	Completed	FALSE	TRUE	TRUE	1	0.2	0
5	Completed	TRUE	FALSE	FALSE	1	18	0
6	Connection	FALSE	TRUE	TRUE	112	353.12	14
7	Construction	FALSE	FALSE	TRUE	1	27	0
8	Failed	FALSE	TRUE	TRUE	36	397.6	0
9	Failed	TRUE	TRUE	TRUE	100	1196.075	0
10	Planning	TRUE	TRUE	TRUE	69	230.59	34

Projects without capacity estimates were assumed zero. From inspection these tended to be improvement projects rather than fresh new capacity.

We take projects that have 'revised' status and have been delivered a planning decision as dropouts (#9). They will have started a new application. We track this instead. This prevents the planning duration estimate being unduly influenced by decision timelines that actually reflect multiple revisions to an application.

The number of projects that failed to proceed from the Planning stage is 100, worth 1196 MW. This is inclusive of revised applications.

The number of projects that failed to proceed from the Connection stage is 36, worth 746 MW.

Note some data quality issues are captured in the table.

#2 1 project was marked as operational despite having no operational timestamp

#4 1 project was marked as operational despite having no construction or operational timestamp.

#5 1 project was marked as operational despite having no planning granted date

Out of those 421 projects that are in the database, and minus 69 planning pending, there are 352. Of those 100 didn't leave planning/ progress from planning *for any reason*.

This gives a crude $100 / 352 \approx 28.41\%$ dropout or non progression rate for the planning stage

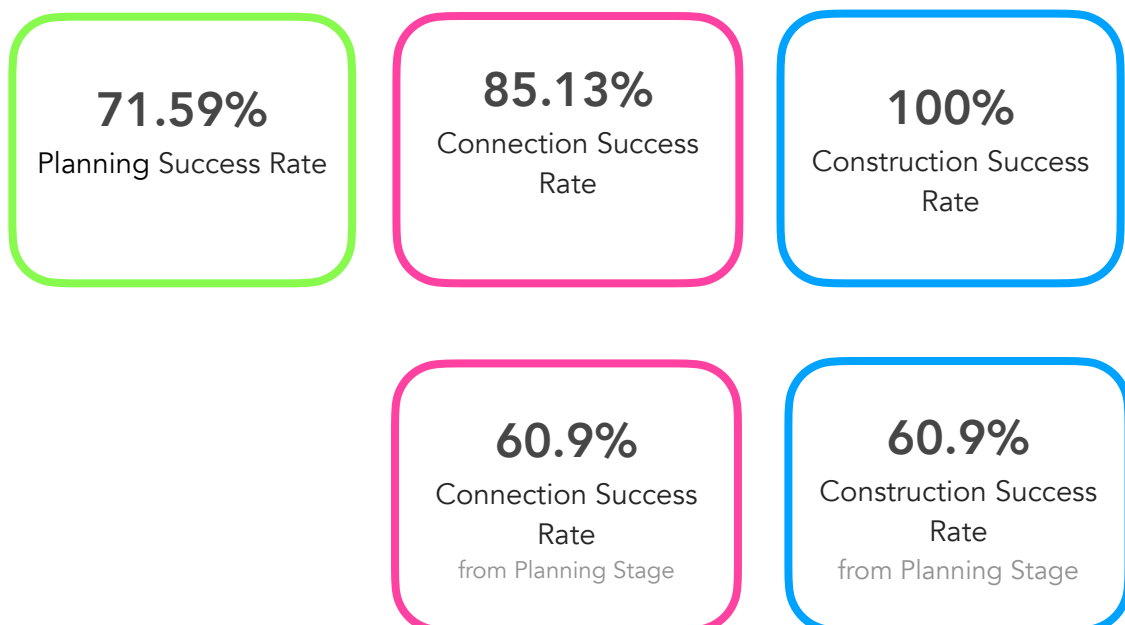
Out of those 252 projects that did proceed from planning, minus the 10 currently in the connection phase, leaves 242 projects. Including those 6 that subsequently revised their application and therefore didn't complete the connection phase, and those projects with expired project granted status enforced by our new ruleset, there are 36 projects that didn't complete connection *for any reason*.

This gives a crude $36 / 242 \approx 14.87\%$ dropout or non progression rate for the Connection stage

Of the remaining 104 projects, 1 is under construction.

Of those 103 that completed the connection phase, 103 became operational. This implies the construction stage has a **0% dropout rate** or a **100% conversion rate**.

Note the milestone timestamp for moving between the connection phase and construction phase is the construction started timestamp. No connection finished or application success flag or timestamp is included in the REPD. A potential weakness of our analysis.



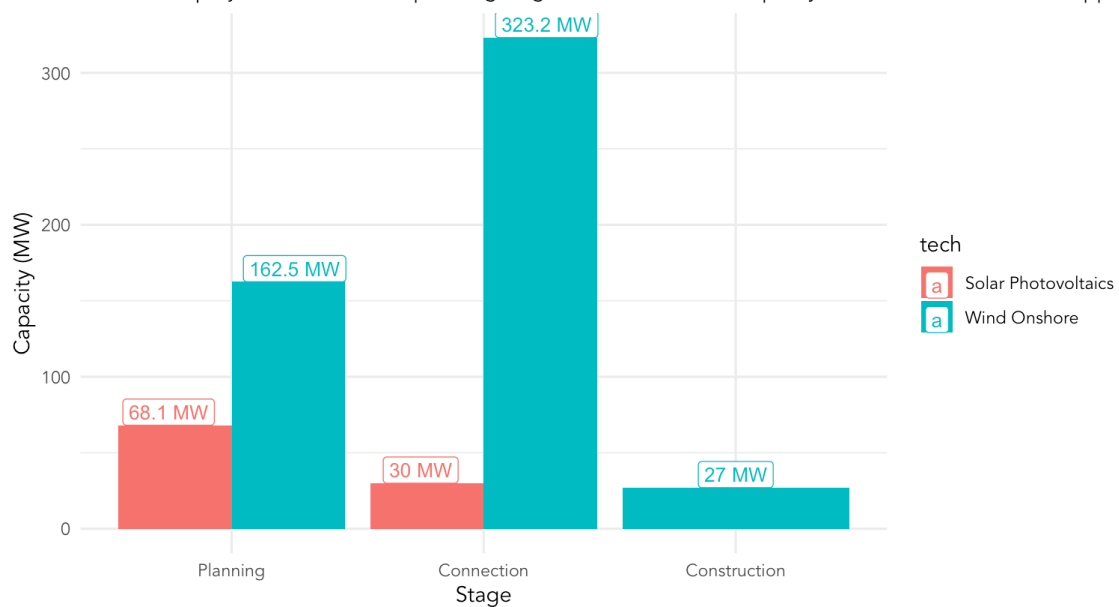
Current Project Development Pipeline

Completed projects were discarded, as estimates of current renewable generation are given by NISRA. The remaining pipeline projects are given by stage of planning and technology.

#	Tech	Current Status	N	Capacity (MW)	# Missing Capacity	Average Capacity (MW)	Padded Capacity (MW)
1	Solar Photovoltaics	Planning	14	68.09	4	6.809	95.326
2	Solar Photovoltaics	Connection	10	29.95	5	5.99	59.9
3	Solar Photovoltaics	Failed	18	198.4	0	11.02	198.4
4	Solar Photovoltaics	Completed	21	209.3	0	9.96	209.3
5	Wind Onshore	Planning	55	162.5	30	6.5	357.5
6	Wind Onshore	Connection	102	323.2	9	3.47	354.44
7	Wind Onshore	Construction	1	27	0	27	27
8	Wind Onshore	Failed	118	1395.27	0	11.824	1395.27
9	Wind Onshore	Completed	82	1195	0	14.57	1195

Capacity at each stage of development process

Includes all projects with status implicating stage. See notes on data quality and treatment of revised applica



The new graph numbers with respect to earlier data.

Moving 'Revised' applications to be explicitly 'Failed' reduced the planning capacity by 319.8 MW 254.4 MW, all among wind.

Applying the new 'Permission Expired' ruleset moved 130 MW worth of expired projects from the 'Connection' stage.

Applying the new 'Application Lapsed' ruleset removed 143 MW worth of projects from the planning stage.

A reminder of the ruleset and the order of application. **Mapping** is applied first and overwritten, by the next rules, the next rules **Permission Expired - Failed, Application Lapsed - Failed**.

2) Permission Expired - Failed

Planning Permission Granted + 6yrs < 2025-08-31 - 2yrs
&
Development Status (short) == Awaiting Construction

2) Application Lapsed - Failed

Planning Application Submitted < 2025-08-31 - 5yrs
&
Development Status (short) == Application Submitted

1) Mapping

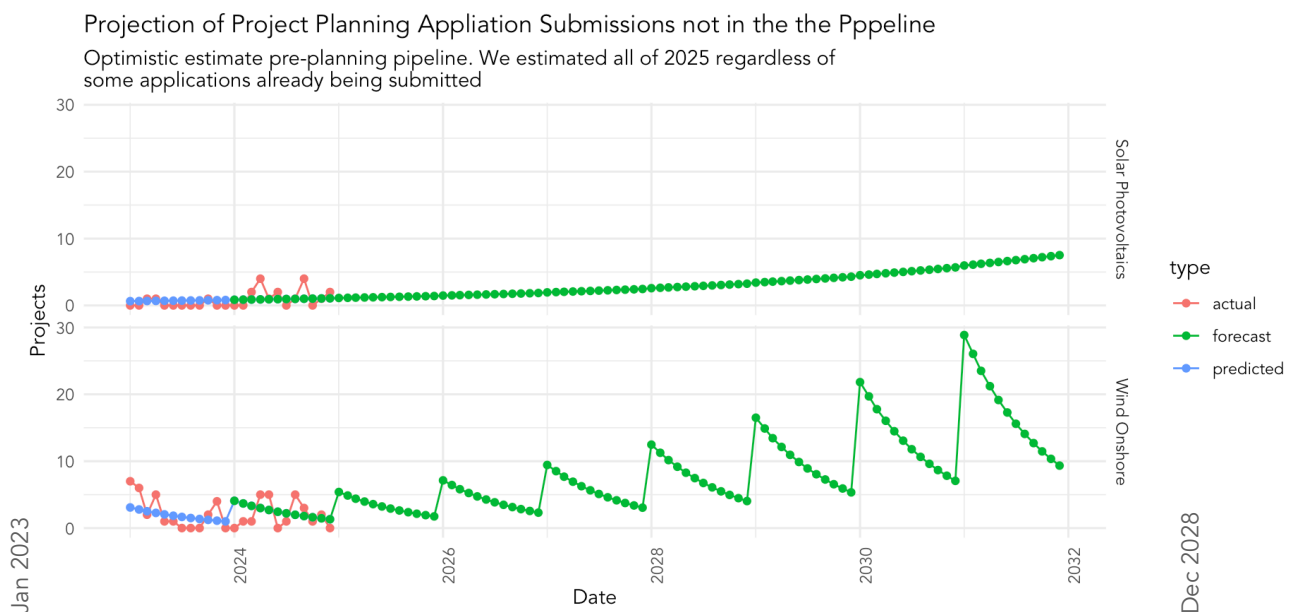
See mapping table in above discussion

Pre - Planning

There is an acknowledgement that the current pipeline will be insufficient to meet the 80% by 2030 target

An estimate of those projects in the *pre-planning* phase, and not available in the planning database is necessary.

Optimistic Pre-planning Projections



```
pd <- glm(data = data, formula = n ~ month*tech + year, family = 'quasipoisson')
```

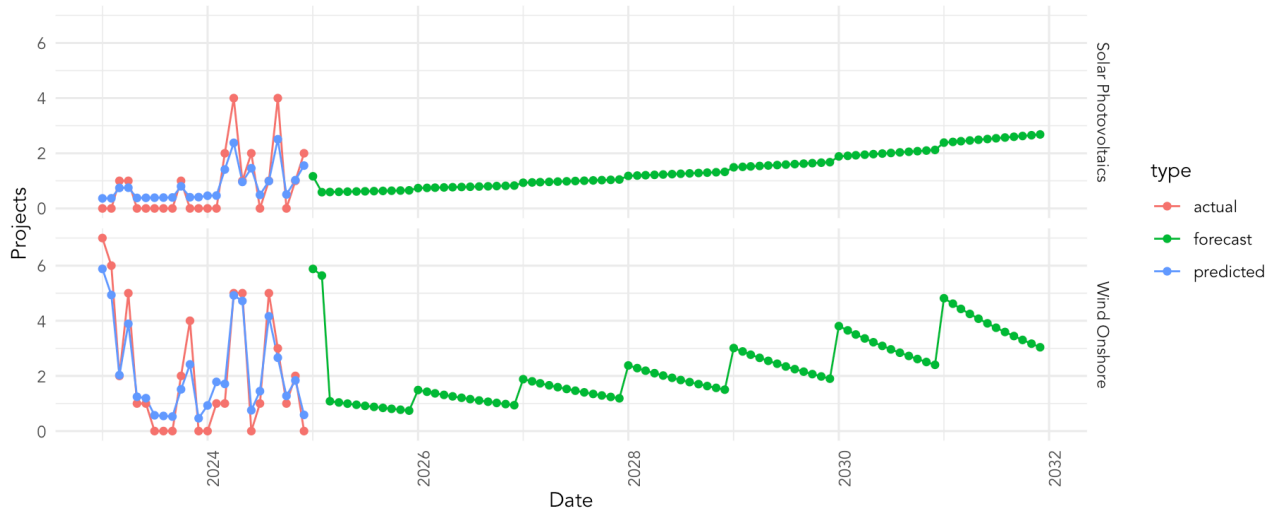
This shows a rapidly accelerating increase in project development rate (Date is planning application submission) for both Solar Photovoltaic and Onshore Wind.

This is the most optimistic scenario for acceleration of renewables. It presumes that development will ramp up gradually with the introduction of a renewable support scheme.

Reasonable Pre - planning Projections

Projection of Project Planning Application Submissions not in the the Pppeline

Conservative estimate pre-planning pipeline. We estimated all of 2025 regardless of some applications already being submitted



```
pd <- glm(data = data, formula = n ~ month*tech+year+offset(log(n+1)), family = 'quasipoisson')
```

This shows a more conservative approach to estimating projects in the pre-planning stage that will submit a planning application (plotted as Date in the x-axis). We treat this as business as usual. It shows a increasing trend year on year and is trained to anticipate the intra-year seasonality dynamics down to a month. It is disaggregated by technology as above

This is the scenario we use below for 2030 estimation of renewable adoption.

Sampled, or Surveyed Pre-planning Projections

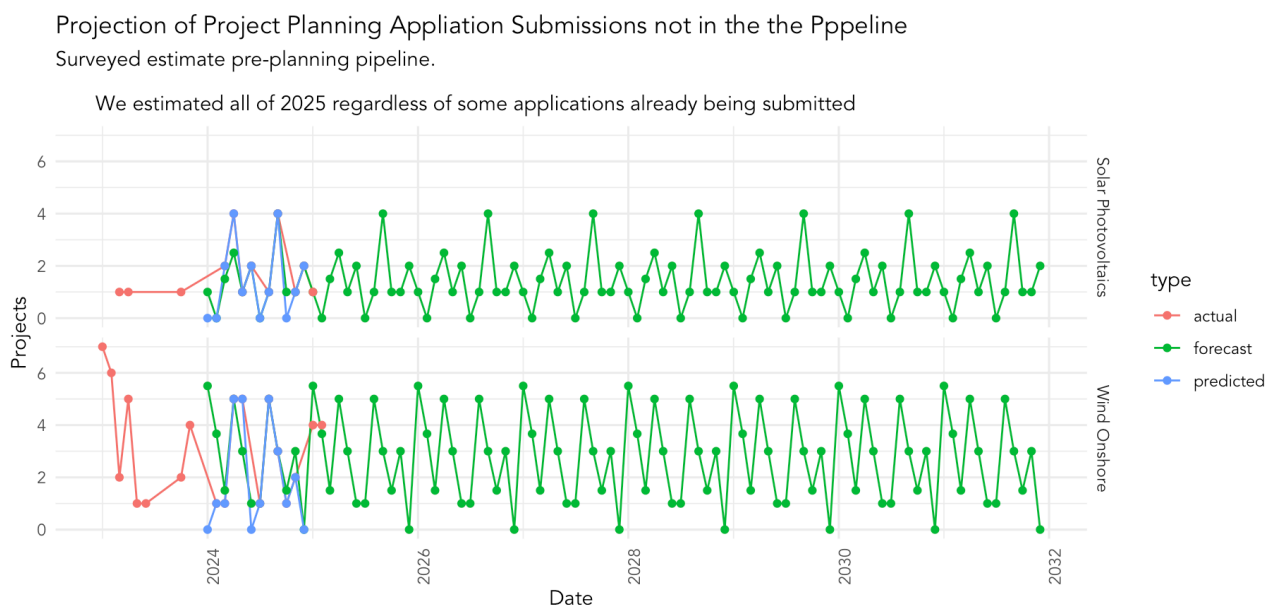
Each pre-planning scenario estimates the number of projects submitting planning applications. The attributes of each new project are sampled from the planning database. This presumes that the new generation of RES will be similar in make up to historical ones across projects attributes.

In the last pre-planning scenario we commit to this assumption, and simply estimate that monthly new projects going forward are the average of the last 2 years new projects for that month.

Note the model is stochastic, if a decimal figure is reported, e.g. 2.6 projects, the interpretation is that the 3 projects are sampled from that month 6/10 times and 2 are sampled 4/10 times for each

of our simulation runs, of which we run multiple. See the the discussion on the model and *nruns* parameter.

Below is the forecasts for the sampled pre-planning estimate scheme. It is made available as a baseline, or no change case. Its yearly new projects do not grow or diminish.



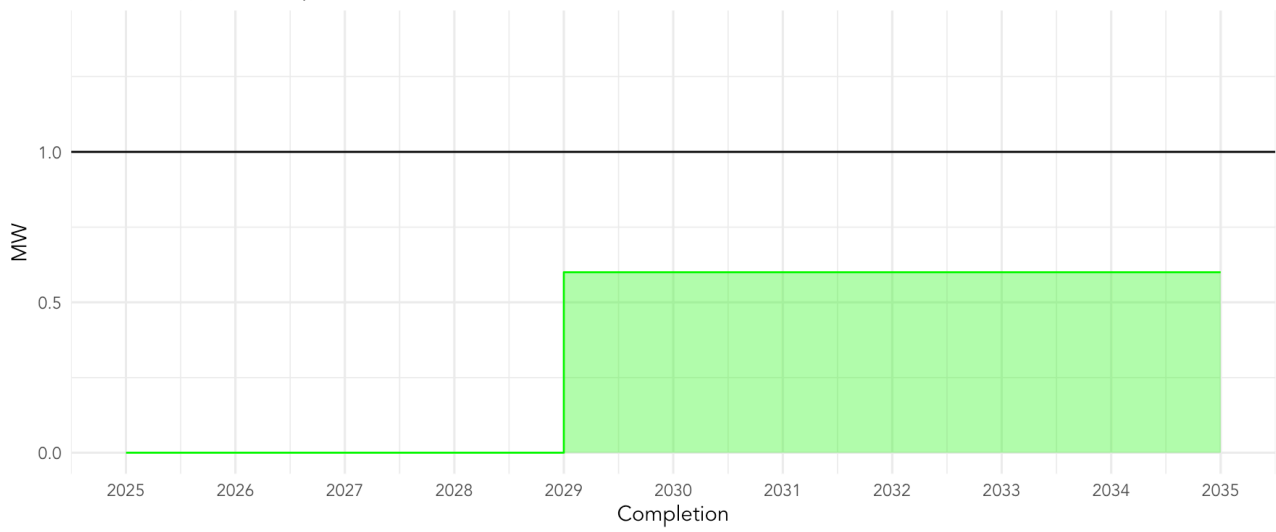
OffShore Wind

Offshore wind is considered to be part of the pre - planning pipeline, as no concrete plans are in place to action this. There is desire to have 1GW in place by 2030. It is not considered in our estimation scheme however. It is made available in addition to RES supply by specifying point estimates of capacity and a point in time, as in the graphic below.

Most all treatments of 80% by 2030 include offshore wind, despite acknowledgement any offshore capacity will arrive later, expected earliest by 2034. Commonly then estimations include a *risk weighted* capacity in calculation. A 10% probability of delivering 1GW of capacity by 2030 constitutes a 0.1GW figure to be included in the estimation.

We have not included any offshore wind estimate in our reported calculation.

Projected capacity estimates of offshore wind
Offshore only. Absolute point estimate



timestamp: 2025-09-01 11:23:22.043677 bootstrap: 50

Model

Results

These are the results for our base model. This took a conservative approach assuming business as usual. For tuning the input parameters see the next section.

We ran two models because of the slightly different treatment the development stages the existing pipeline was distribution across. We were able to assume that projected projects would all start from the planning stage.

So we projected projects in the pre-planning phase that we anticipate bringing forward a planning application at some time from start 2025 through to the end of 2028.

Configuration

Baseline Renewable 2025	3,961 GWhr
Total Energy Requirement 2030	9 TWhr
Capacity Factor	22%
Pre-planning	Conservative
Offshore Wind	FALSE
Date	NA
Capacity	NA
Stage Progression Probablilty	Empirical
Stage Progression Duration	Empirical

Statistical Notes

We run multiple instances of a simulation to generate uncertainty intervals and smooth out distributions. Our *nrun* parameter determines the number of runs. Higher is more rigorous, but takes longer. Publication and reporting quality results should use minimum 80.

It is good practice to append run parameters to result sets.

- Skewed distribution of data required >> samples for bootstrapped confidence interval
- Distributions of MW were right skewed
- Distributions of MW were long tailed

Shortcomings

If a project is in the pipeline already we take it to be at the start of the stage rather than at some point of progressing through the stage.

We will have double counted some early 2025 projects taking the current pipeline up to Q2 2025, and our projected projects from the start of 2025.

Capacity Factor should be stratified for technology types.

As per other analyses, we only consider Solar PV, Offshore and Onshore Wind.

Data Quality

We rely on supplied data sets for data quality. Though we do note discrepancies.

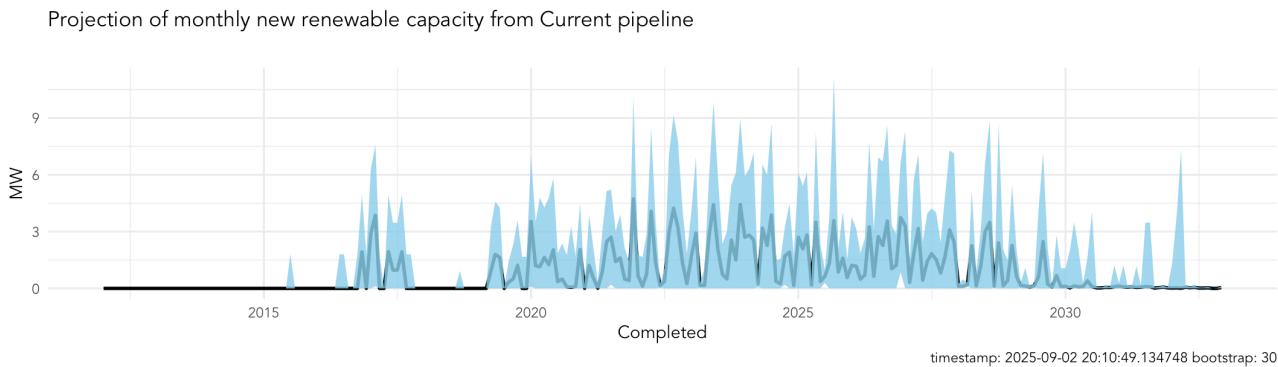
Some projects are being reported as status: *awaiting construction*, despite their planning permission having expired and a other projects reporting a status as status: *planning expired* (e.g. ref: 4251)

Update 01/09/25: We have introduced a new ruleset that excludes the majority of these cases

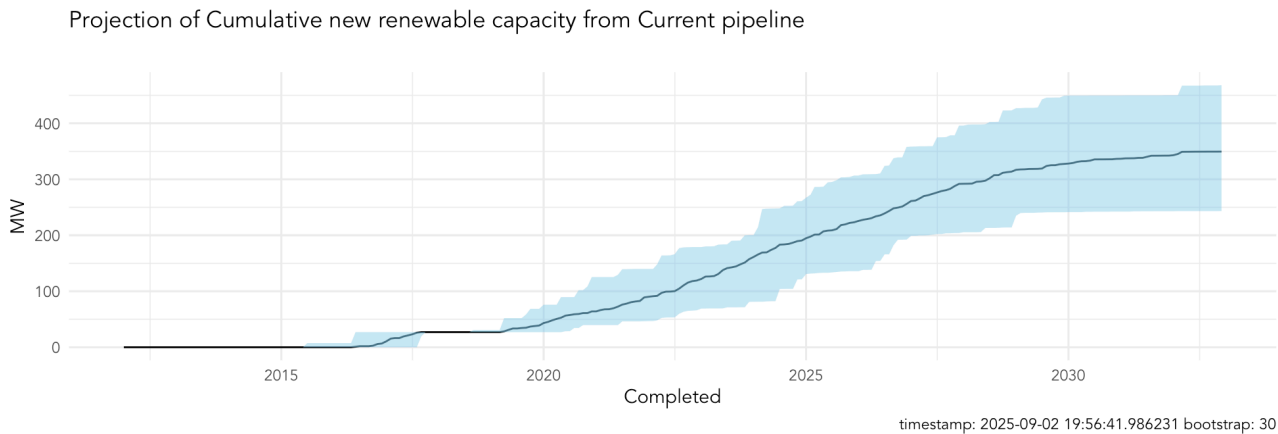
Some projects are missing Installed MW capacity figures. These were left blank as as manual inspection implied some of these were upgrade or refurbishment proposals with existing MW capacity and uncertain extra capacity.

Current Pipeline

Monthly



Cumulative

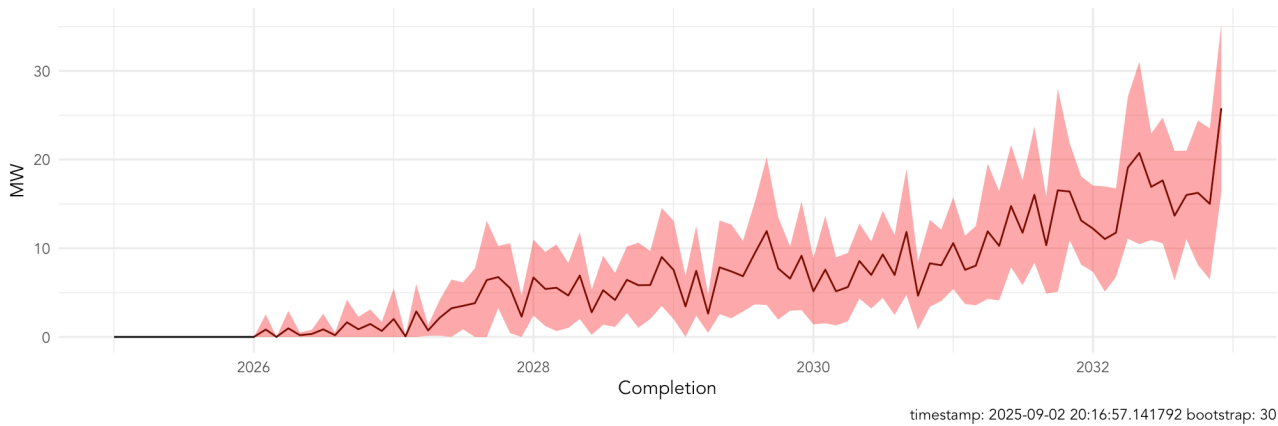


It is noted that projects in the current pipeline may have start dates before our current date due to the stochastic nature of our modelling. It is further noted that the these projects are substantial. Please consult the data quality notes.

Pre-planning Projects

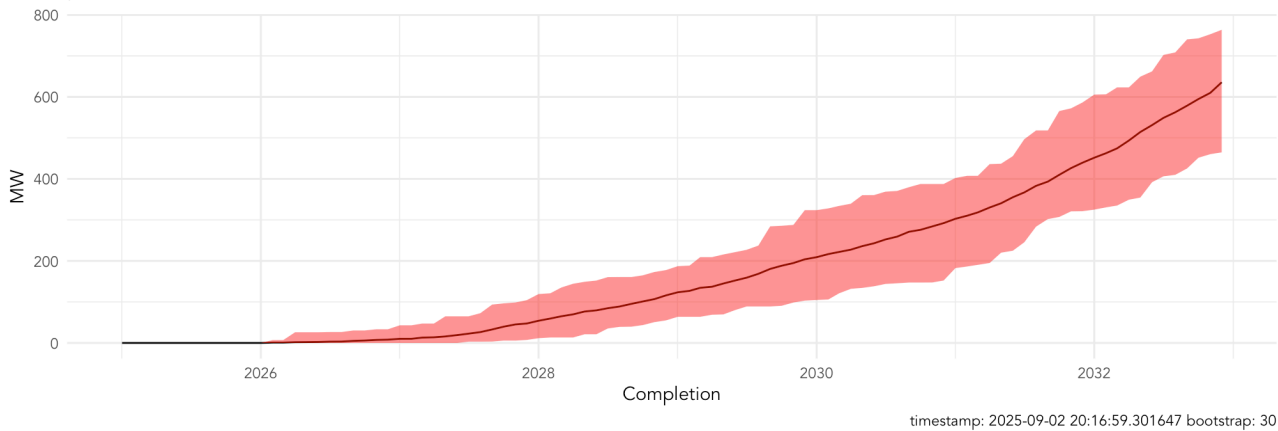
Monthly

Projected Monthly new renewable capacity from projects in pre-planning



Cumulative

Projected Cumulative new renewable capacity from projects in pre-planning



303 MW

New Capacity from
Projected Pipeline
by 2030

336 MW

New Capacity from Current
Pipeline
by 2030

639 MW

Total New Renewable
Capacity
by 2030

2279 MW

800 MW + 1640 MW

Total Renewable Capacity inclusive
of current capacity by 2030

4151 MW

8 TWhr/yr
capacity factor of 22%

Total Capacity Requirement by 2030

54.9%

% Renewable by 2030
Target Miss

Interactive Application

User Interface

Inputs on the left, graphics in the middle, high level metrics to the right

Blue sliders require rerunning the simulation

Green sliders or other inputs only affect output metrics and update the last computed result set.

Some UI elements are conditional on others. Duration for example has two configurations: Use empirically derived stage progression durations, or use your own, for which you may chose the durations for *Planning*, *connection* and *Construction*. While you can't specify these to be empirical or custom individually, the default values are the empirically derived values.

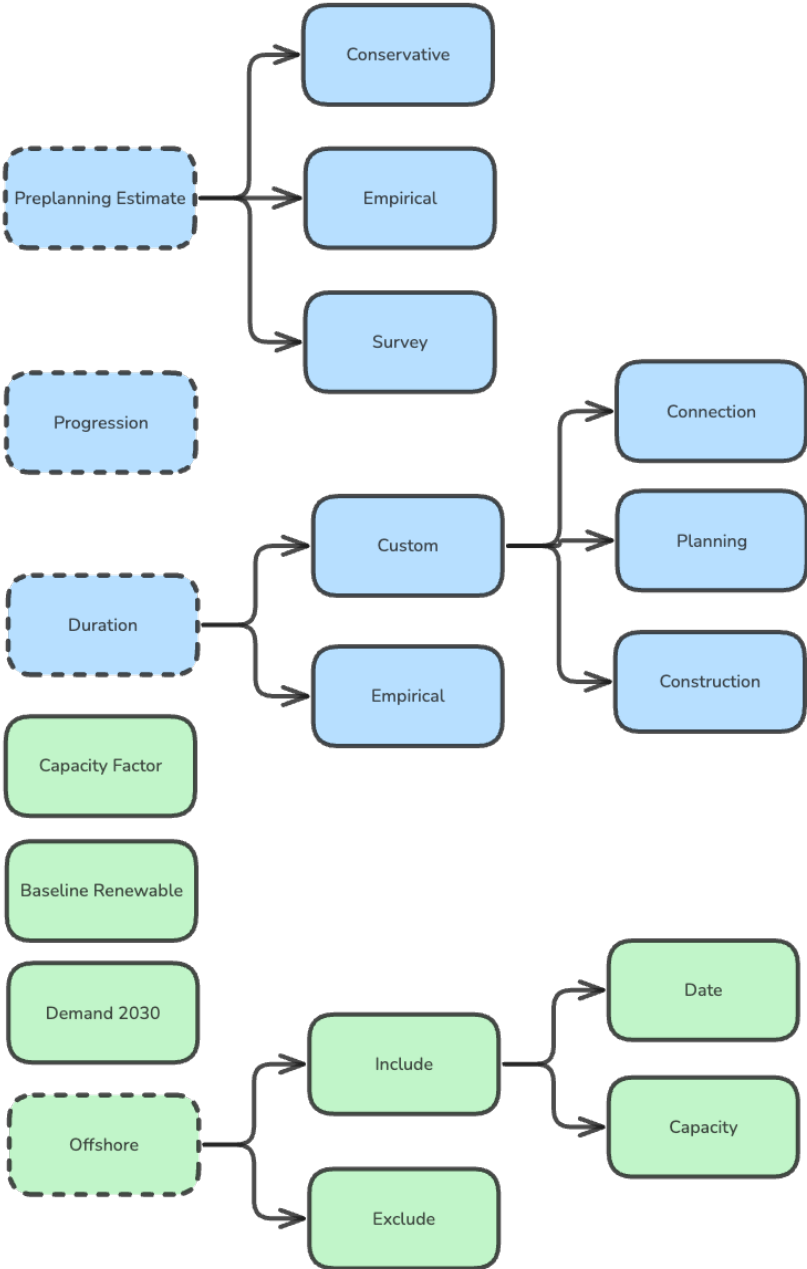
Take care when specifying the offshore wind input, the inclusion button must be clicked to be included, and the date must be 2030 or earlier to be included in the 2030 target figures on the left.

Model

The model is a discrete event simulation using Monte Carlo simulations to build bootstrapped confidence intervals and quantify uncertainty by repeat sample from distributions.

It takes some time to run. The *nruns* parameter is a model meta parameter that describes the number of repeat simulations to run. It is surfaced to model users to decrease to experiment and get broad strokes from model outputs. It is vital to statistical relevance of both the point estimate and the confidence intervals that this *nruns* is increased for comparison of scenarios, reporting, publication and sensitivity analysis.

Inputs Schematic of Cascading User Input



Appendix

Energy Scenario Platforms

Energy Savings and Impacts Scenario Tool (ESIST)

<https://www.epa.gov/statelocalenergy/energy-savings-and-impacts-scenario-tool-esist>

Climate Scenario Risk Tool from the World Business Council

<https://climatescenariocatalogue.org/explore-the-data>

Local Authority in Scotland scope seems similar to ours but implementation unclear

<https://edinburghcentre.org/projects/carbon-scenario-tool-pathfinder-project>

Canada Scenario Modelling Tool - very cool - 4 tools

<https://cme-emh.ca/en/open-source-tools/visualization-tools>

Commercial Grade Software from S&P

<https://www.spglobal.com/commodity-insights/en/products-solutions/carbon-scenarios/energy-scenarios>

Data Sources

All Ireland Adequacy Assessment - SONI/ EirGrid

NISRA

Aurora Report

Solar photovoltaics deployment

Monthly deployment of all solar photovoltaic capacity in the United Kingdom.

<https://www.gov.uk/government/statistics/solar-photovoltaics-deployment>

Planning Portal

<https://planningregister.planningsystemni.gov.uk/simple-search>

Renewable Energy Planning Database - REPD - NI

List of planning applications for single wind turbines, wind farms and solar farms by status, April 2002 to March 2025.

<https://www.infrastructure-ni.gov.uk/publications/renewable-energy-planning-applications-list-april-2002-march-2025>

Renewable Energy Planning Database - REPD - UK

<https://www.gov.uk/government/publications/renewable-energy-planning-database-monthly-extract>

Commissioned Projects database

Resources

Climate Change Committee

<https://www.theccc.org.uk/publication/adapting-to-climate-change-progress-in-northern-ireland/>

Design considerations for a renewable electricity support scheme for Northern Ireland: response

<https://www.economy-ni.gov.uk/publications/design-considerations-renewable-electricity-support-scheme-northern-ireland-response>

Accelerating Renewables in Northern Ireland

KPMG - RenewableNI

<https://renewableni.com/wp-content/uploads/2023/09/RNI-Report-Accelerating-renewables-in-Northern-Ireland-online-version.pdf>

Aurora

Accelerating Renewables in Northern Ireland

A High level design of Renewable support scheme

<https://www.economy-ni.gov.uk/sites/default/files/publications/economy/Aurora-accelerating-renewables-support-scheme-HLD.pdf>

NIRO

<https://www.economy-ni.gov.uk/sites/default/files/publications/economy/NIRO-how-it-works.pdf>

DfE - road to net zero

<https://www.economy-ni.gov.uk/publications/energy-strategy-path-net-zero-energy>

DfE, NISRA Energy in Northern Ireland

<https://www.economy-ni.gov.uk/sites/default/files/publications/economy/Energy-Northern-Ireland-2024-Report.pdf>

References

NISRA

https://datavis.nisra.gov.uk/Economy/electricity-consumption-and-renewable-generation-report.html#2_Electricity_Consumption_from_Renewable_Sources_-_Monthly_data

Renewable Energy Support Scheme (RESS)

<https://www.economy-ni.gov.uk/sites/default/files/publications/economy/Renewable-Electricity-Support-Scheme-response.pdf>

KPMG RenewableNI - Economic Review of Small Scale Wind

Reports a capacity factor of ~22% for small scale wind

<https://renewableni.com/wp-content/uploads/2021/06/An-Economic-Review-of-Small-Scale-Wind34.pdf>

Undertaken for and on behalf of RenewableNI

Aurora

Page 80 reports a load factor of 23% on average for onshore renewable projects

Glossary

RES - Renewable Energy Source

RES-E stands for Renewable Energy Sources – Electricity. It is used to distinguish from other sources used for heat, or transport, RES-H and RES-T, for example.

Dispatchable Generation - Sources of electricity that can be used on demand and dispatched at the request of power grid operators according to needs
Does not include wind and solar generation which are non-dispatchable generation.

Capacity

GW is an energy per unit time measure. It is used to estimate ability to accommodate peaks in demand

Generation

Generation is referred to over time period. Typically a month or year and given in TWh.

Average GW from TWh/y

$$\text{GW (avg)} = \frac{\text{TWh per year}}{8.76}$$

(Leap year: divide by 8.784.)

Note the difference between nameplate and average measures of capacity and generation, below.

Capacity Factor

Generators will not produce at full power for the entirety of e.g. a year. To describe this a percentage metric between full and delivered capacity is calculated called a capacity factor.

For small scale wind installations this is estimates at 22%, meaning over some time period, such as a year the installation is only producing 22% of its possible (nameplate) output.

$$\text{Capacity Factor} = \frac{\text{Actual Energy Generated in period (MWh)}}{\text{Nameplate Capacity (MW)} \times \text{Hours in period}}$$

Load Factor is the same a capacity factor in the majority of cases

Nameplate capacity is the generation capacity under optimal conditions
If you want nameplate capacity at a capacity factor (CF):

$$\text{GW (nameplate)} = \frac{\text{TWh/yr}}{8.76 \times \text{CF}}$$

Example: 3.5 TWh/y @ 40% CF

$$\frac{3.5}{(8.76 \times 0.40)} \approx 1.0 \text{ GW}.$$

Contract for Difference

CfD - a government guaranteed contract providing a mechanism for a set price that electricity can be sold for.

The RESS is to be of this type.

Bid and Auctions

In the context for CfD bids are for a set strike price that sets the guaranteed electricity price.

Auctions are more typically referred to as **Allocation Rounds**.

Offers refer to the offered capacity e.g. 10GW.

Win Rate is the percentage of eligible energy volume that secures a contract

$$\text{Win Rate\%} = \frac{\text{Awarded GWh}}{\text{Eligible GWh}}$$

Procurement Ratio is the required win rate for some target, or other delivery set. Here generally referring to 80% by 2030.

$$\text{Procurement Ratio\%} = \frac{\text{Target GWh}}{\text{Eligible GWh}}$$

Competition Ratio or Oversubscription ratio

$$\text{Competition ratio} = \frac{\text{Eligible volume}}{\text{Awarded volume}} = \frac{1}{\text{Win rate}}$$

If you do not win a contract, you can resubmit given the project is still eligible. Otherwise the project is said to proceed **without support**, or is referred to as a merchant project.

Auctions are usually set a fixed capacity, or generation (MW or MWh) such as 1500 GWh/y over 10 years for example.

Alternatively they may be limited by a price or budget cap.

A smaller pot usually means more competition and higher prices

Auction Criteria for Project Eligibility

Capacity Threshold Typically > 5 MW

Planning Consent & Land Rights Required

Must have a valid Grid Connection Offer

Supply Chain Plan Required for ≥ 300 MW or floating offshore projects

Sealed bid, below admin strike price

Auction Pot

An auction may be divided into pots relevant to RES technologies or other.

Clearing Price

The highest winning bidding strike price that still fits within the pot's capacity.

Also said to be the price at which the auction *clears*.

It has units £/MWh.

Pay per clearing is a type of auction where all winning bidders get paid the auction's clearing price regardless of their individual bid.

An **Admin strike cap** is the ceiling placed on bids. Any projects bidding above this are disqualified.

NIRO - Northern Ireland Renewables Obligation

An initiative discontinued in 2017. Legislation for suppliers to present ROC (Renewable Obligation Certificates)

A **Corporate Power Purchase Agreement (CPPA)** is a long-term contract where a company buys electricity directly from a renewable energy generator, such as a wind or solar farm, rather than from a traditional electricity supplier. It is an example of an alternative route to market for renewable energy projects.

Figures

AIRAA

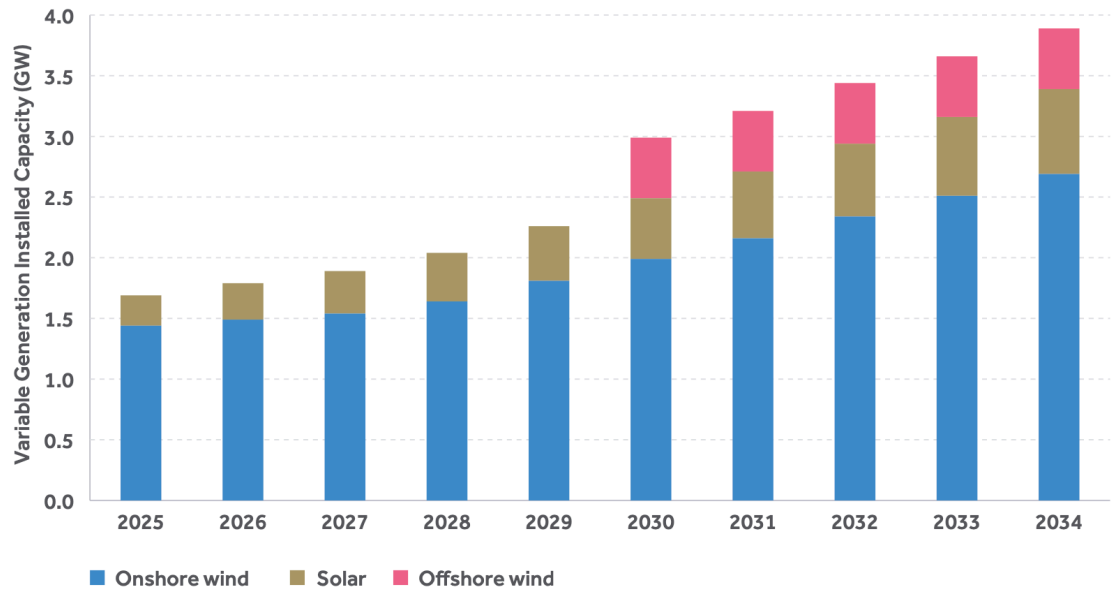


Figure 5.5 Assumed growth of renewable capacity in Northern Ireland

AURORA

Cumulative operational generation volumes successful in first two auctions⁴⁸
GWh/year

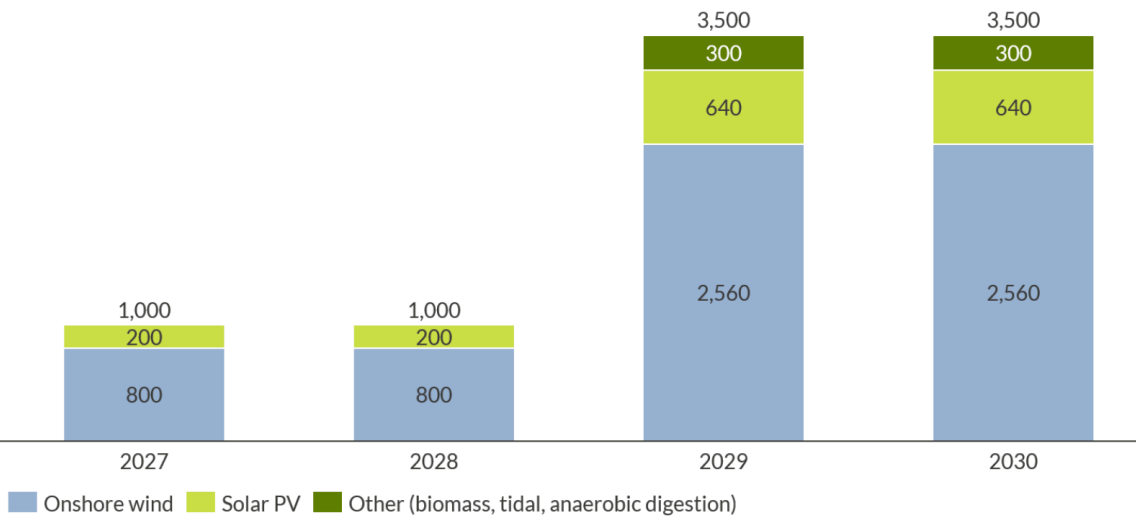
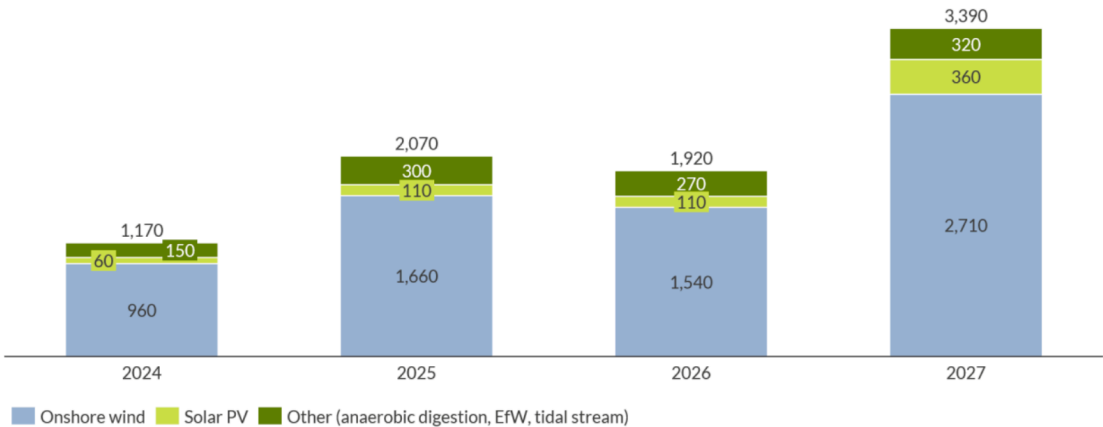


Figure 11: Cumulative operational generation volumes successful in first two auctions

Table 7 – Illustrative outcome of Auction 1

Pot	Technology	Generation (GWh/year)	Generation (% of total)	Capacity (MW)
1	Onshore wind	800	80%	340
1	Solar PV	200	20%	255

Annual generation volumes eligible for support scheme
GWh/year



Capacity eligible for support scheme
MW

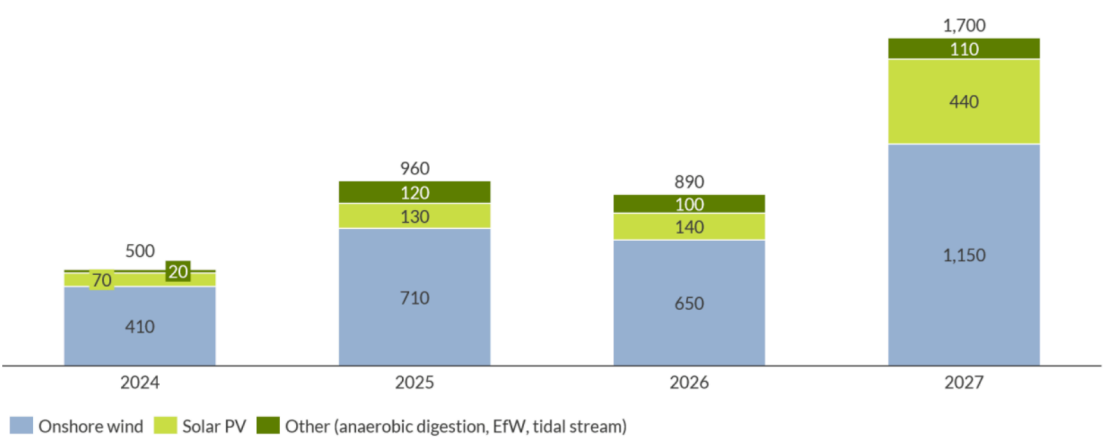
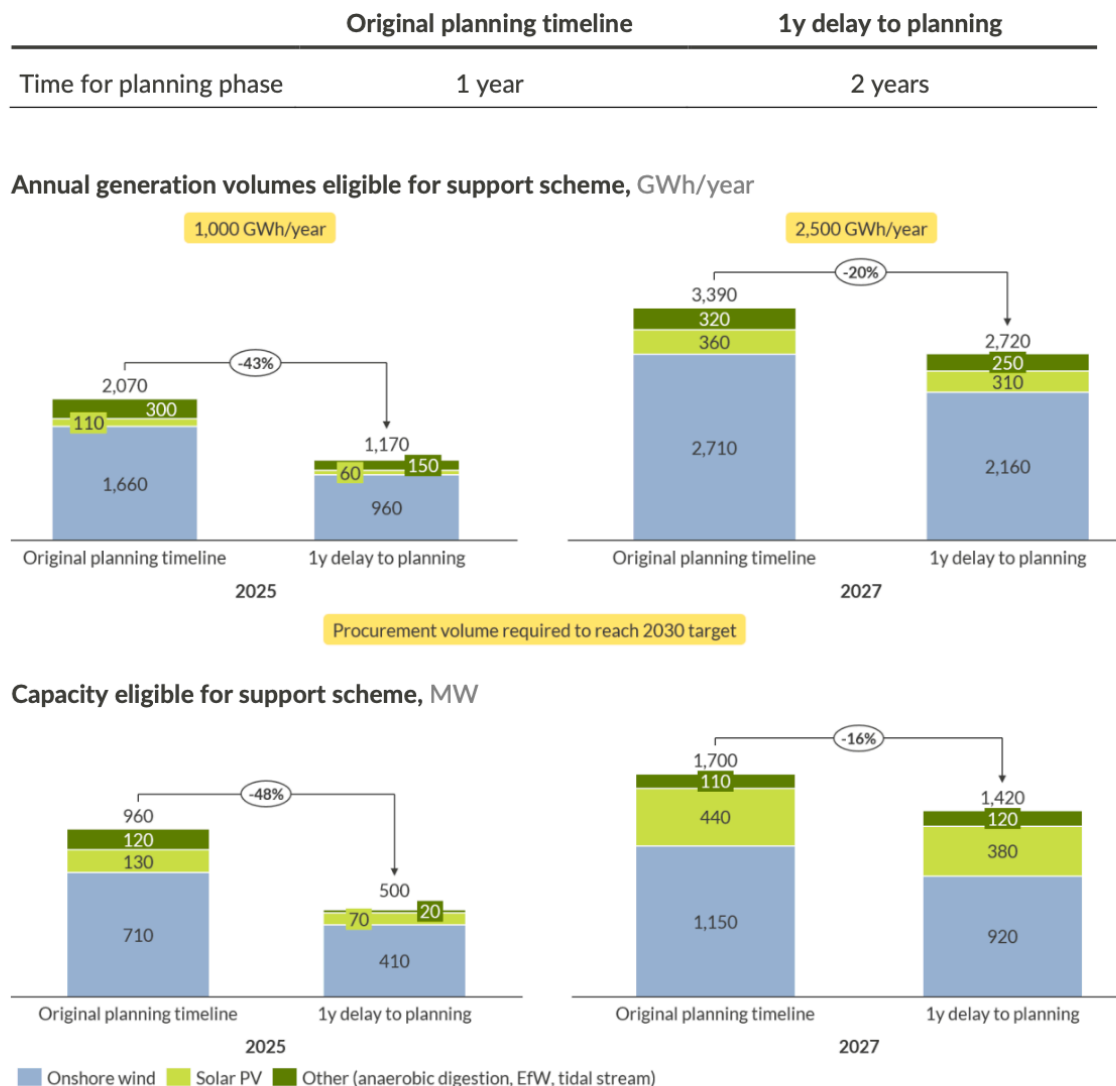


Figure 10: Onshore renewables capacity eligible for the Support Scheme (assuming 1 TWh is successful in 2025)

4.7 Sensitivity on planning timeline

The below chart illustrates the effect of a one-year delay to planning timelines on the capacity and associated production volumes eligible for the Scheme⁵⁴ in the first auction (left) and second auction (right).



Under a one-year delay to the planning timeline, both auctions could still procure sufficient volumes, but with a much lower competition ratio. The first auction would need to procure 85% of eligible volumes (up from 48% in the one-year planning scenario); similarly, the second auction would need to procure 92% of eligible volumes (up from 74%). This could result in greater strategic bidding, and higher strike prices.

All-Island Resource Adequacy Assessment - SONI / EirGrid

Table A3.5 All renewable energy sources in Northern Ireland (MW)

At year end:	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
All Wind	1435	1485	1535	1635	1810	2485	2660	2835	3010	3185
Solar PV	250	300	350	400	450	500	550	600	650	700
All Biomass/ Biogas/LFGas/ WTE	250	300	350	400	450	500	550	600	650	700
	79	79	79	79	79	79	79	79	79	79
Renewable CHP	3	3	3	3	3	3	3	3	3	3
Hydro	6	6	6	6	6	6	6	6	6	6

Table A1.1 Demand forecast for the median scenario in calendar year format for Ireland and Northern Ireland

Median	Calendar Year TER (TWh)						TER peak (GW)			Transmission peak (GW)		
Year	Ireland		Northern Ireland		All-Island		Ireland	Northern Ireland	All-Island	Ireland	Northern Ireland	All-Island
	TER	% growth	TER	% growth	TER	% growth						
2023	34.0		8.09		42.1		5.76	1.6	7.36	5.64	1.57	7.21
2024	35.8	5.3%	8.21	1.6%	44.1	4.6%	5.95	1.59	7.54	5.83	1.56	7.39
2025	37.7	5.2%	8.33	1.4%	46.1	4.5%	6.13	1.61	7.72	6.02	1.57	7.57
2026	39.5	4.7%	8.57	2.8%	48.1	4.4%	6.32	1.67	7.97	6.21	1.63	7.82
2027	41.1	4.1%	9.05	5.7%	50.2	4.4%	6.46	1.75	8.19	6.35	1.71	8.04
2028	42.4	3.1%	9.31	2.9%	51.7	3.0%	6.57	1.79	8.33	6.46	1.76	8.18
2029	43.5	2.8%	9.58	2.9%	53.1	2.8%	6.64	1.84	8.45	6.53	1.80	8.30
2030	44.7	2.6%	9.83	2.5%	54.5	2.6%	6.72	1.87	8.53	6.61	1.84	8.39
2031	46.0	2.9%	10.11	2.9%	56.1	2.9%	6.83	1.91	8.66	6.71	1.88	8.52
2032	47.2	2.7%	10.47	3.5%	57.7	2.8%	6.92	1.95	8.79	6.81	1.91	8.63
2033	48.4	2.5%	10.71	2.3%	59.1	2.5%	7.03	1.99	8.90	6.91	1.96	8.76
2034	49.5	2.4%	10.94	2.2%	60.5	2.3%	7.12	2.04	9.03	7.01	2.00	8.88

Shaping Our Electricity Future Roadmap Version 1.1’ published by EirGrid and SONI in July 2023

Table 12 provides a summary of the major renewable generation capacities assumed to be in place in Ireland and Northern Ireland by 2030.

Table 12: Summary of assumed renewable generation capacities in 2030		
Source	Ireland (MW)	Northern Ireland (MW)
Onshore wind	9,000	2,450
Offshore wind	5,000 (+ 2,000 MW assigned to hydrogen production)	500
Solar ⁹	8,000	600
Total	22,000	3,550

“The subsequent revised legal target of 80% RES-E in Northern Ireland by 2030 led to a recognition that offshore generation capacity would need to be delivered by 2030 to help achieve the target.”

Updates from first version

Pipeline state

Those 'revised' moved to failed explicitly - all will not have met our new criteria below anyway. They will almost have always started a fresh application. Note the Database natural unique identifier is on a project application basis, not project. This means the primary key is the application reference, and failure in this context means failure to proceed, not whole project failure.

Filter

1) Those projects that have a planning permission date expired up to 2 years ago are excluded. Many of these projects were observed to still have a status of 'Awaiting Construction'. Planning expires about 5 years from grant. The leeway is that some projects have a planning permission expired date, but have proceeded to construction. It may reflect a delay in updating the dataset on the part of the maintainers.

2) Those projects that have the 'Planning Submitted' date more than 5 years in the past are marked as failed. Many of these are marked as 'Planning Permission Submitted'. It is taken as unreasonable that even with appeals, that these are actively being pursued.

For the purpose of fixing the time for consistent analysis 'today' refers to
Sys.Date()
"2025-08-31"

`Planning_expired < as.Date("2025-08-31") - 2*365`

`Planning < as.Date("2025-08-31") - 5*365`

The impacts on estimates include diminished estimates of our planning pipeline and also on our estimates of the pre-planning pipeline, as we model the latter on the former.

Advantages include not including suspect projects in our current pipeline that by most reasonable estimates of progression and given their start date, should have finished in the past. While not wholly unreasonable given the stochastic nature of our modelling, the sheer load of projects completing in the past indicates data quality at source.

Offshore Wind

Offshore wind estimates can now be included on a risk weighted basis (i.e. less than the anticipated 1GW) on an arbitrary date (year-month) that may be after 2030.

Horizon

We model further ahead to include the projections of all metrics past 2030.

Changelog

Removed from intro we now consider pre-planning through modelling schemes and scenario analyses

"We assume that all projects pertinent to new 2030 supply are already in the planning or development or commissioning process."

Updated **graphics** and **transition probabilities**.

Updated Ruleset